

## Example and Exercise 1

| Item                | Example description                                                                         |
|---------------------|---------------------------------------------------------------------------------------------|
| Object              | Car                                                                                         |
| Possible attributes | brand, model, colour, speed, fuelLevel, engineOn                                            |
| Possible behaviours | startEngine(), stopEngine(), accelerate(), brake(), turnLeft(), turnRight()                 |
| Context decision    | A simple Week 1 model may omit tyres, engine capacity or GPS unless the program needs them. |

## Student worksheet

| # | Favourite object | Relevant attributes (what it has)                                                                | Behaviours (what it can do)                                                                                                                          |
|---|------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Analogue Watch   | int currentHour, int currentMinute, int currentSecond, String currentDate, boolean isCrownPulled | tick(), advanceDate(), calculateSecondAngle(), calculateMinuteAngle(), calculateHourAngle(), pullCrown(), pushCrown(), setTime(int hour, int minute) |
| 2 | Phone            | boolean isPoweredOn, int batteryLevel, boolean isLocked, int volume, boolean isOnCall            | powerOn(), powerOff(), chargeBattery(int amount), unlock(String passcode), adjustVolume(int level), makeCall(String number), endCall()               |
| 3 | Digicam          | boolean isPoweredOn, int zoomLevel, int batteryLevel, int storageUsed, boolean flashOn           | powerOn(), powerOff(), zoomIn(), zoomOut(), chargeBattery(int amount), capturePhoto(), recordVideo(), deletePhoto(), toggleFlash()                   |
| 4 | Headphones       | int batteryLevel, int volumeLevel, boolean isConnected, boolean                                  | chargeBattery(int amount), adjustVolume(int level), connect(String deviceName), disconnect(),                                                        |

|   |       |                                                                                                  |                                                                                                    |
|---|-------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
|   |       | noiseCancellingOn, boolean<br>isPlaying                                                          | toggleNoiseCancelling(), playAudio(),<br>pauseAudio()                                              |
| 5 | Pasta | int quantity, double<br>cookingTime, boolean isCooked,<br>boolean isDrained, String<br>sauceType | boil(int minutes), checkCookingTime(),<br>drain(), addSauce(String sauceType),<br>serve(int grams) |